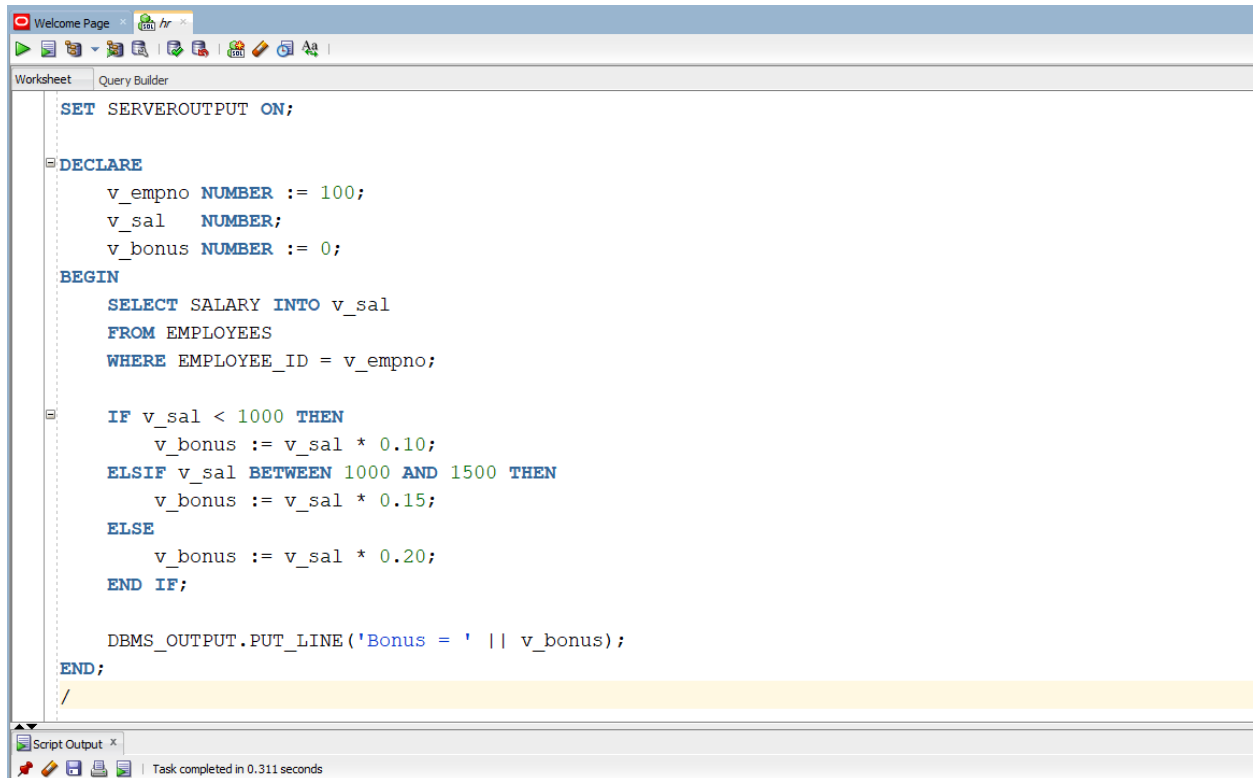


DBS LAB 10

In Lab Tasks

Task 1:



The screenshot shows a web-based interface for writing and executing PL/SQL code. The main window is titled 'Query Builder' and contains the following code:

```
SET SERVEROUTPUT ON;

DECLARE
    v_empno NUMBER := 100;
    v_sal    NUMBER;
    v_bonus  NUMBER := 0;
BEGIN
    SELECT SALARY INTO v_sal
    FROM EMPLOYEES
    WHERE EMPLOYEE_ID = v_empno;

    IF v_sal < 1000 THEN
        v_bonus := v_sal * 0.10;
    ELSIF v_sal BETWEEN 1000 AND 1500 THEN
        v_bonus := v_sal * 0.15;
    ELSE
        v_bonus := v_sal * 0.20;
    END IF;

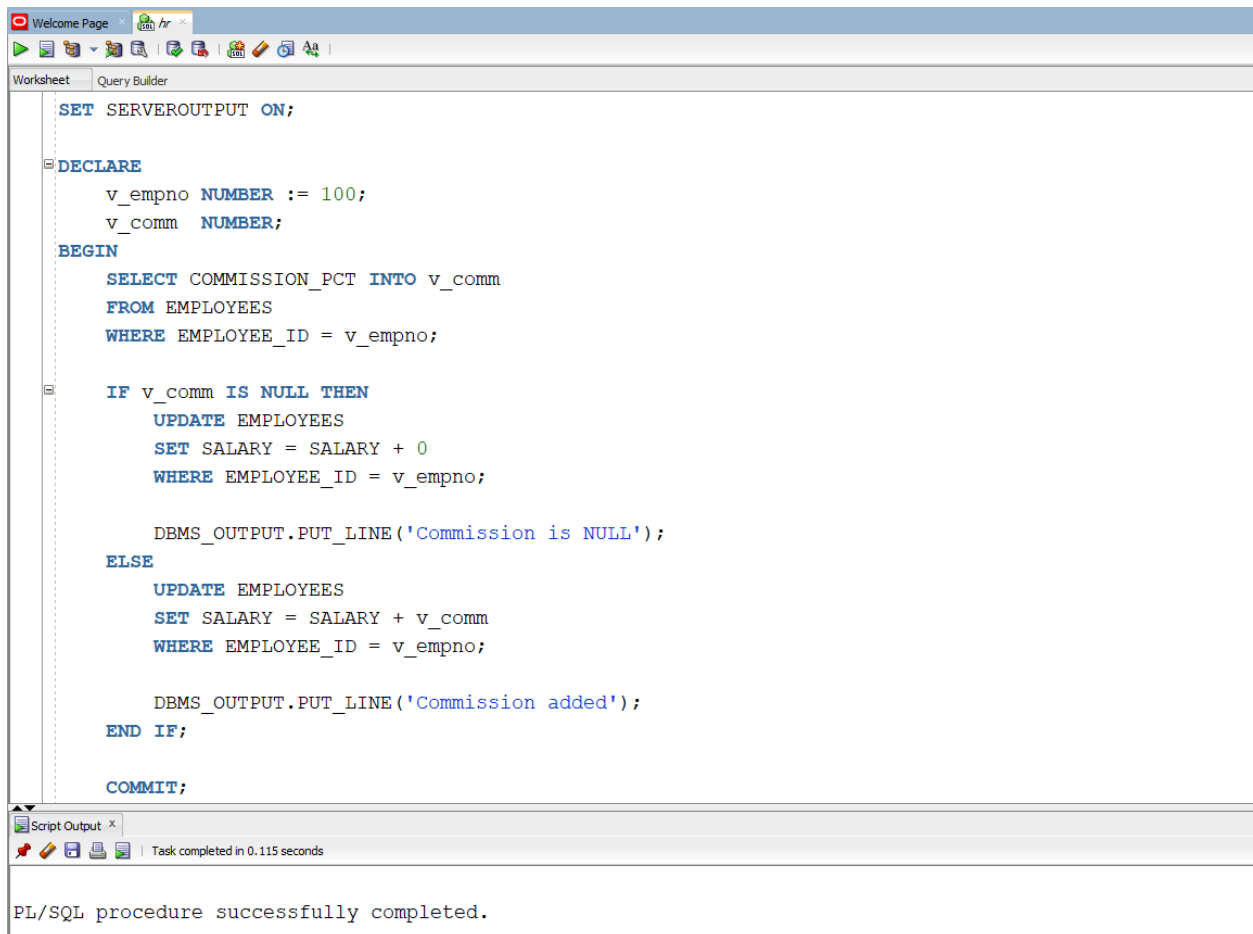
    DBMS_OUTPUT.PUT_LINE('Bonus = ' || v_bonus);
END;
```

Below the code editor, there is a 'Script Output' window showing the result of the execution:

```
PL/SQL procedure successfully completed.
```

The interface includes a top toolbar with various icons for file operations and a bottom status bar indicating 'Task completed in 0.311 seconds'.

Task 2:



The screenshot displays the Oracle SQL Developer environment. The main window, titled 'Query Builder', contains a PL/SQL script. The script starts with 'SET SERVEROUTPUT ON;', followed by a 'DECLARE' section with two variables: 'v_empno' of type 'NUMBER' set to 100, and 'v_comm' of type 'NUMBER'. The 'BEGIN' block contains a 'SELECT' statement to fetch 'COMMISSION_PCT' into 'v_comm' from the 'EMPLOYEES' table where 'EMPLOYEE_ID' is 'v_empno'. An 'IF' statement checks if 'v_comm' is NULL. If true, it updates the 'SALARY' by adding 0. If false, it updates the 'SALARY' by adding 'v_comm'. Both paths use 'DBMS_OUTPUT.PUT_LINE' to output a message ('Commission is NULL' or 'Commission added'). The script ends with 'END IF;' and 'COMMIT;'.

Below the script editor, the 'Script Output' window is visible, showing the message: 'Task completed in 0.115 seconds' and 'PL/SQL procedure successfully completed.'

```
SET SERVEROUTPUT ON;

DECLARE
    v_empno NUMBER := 100;
    v_comm  NUMBER;
BEGIN
    SELECT COMMISSION_PCT INTO v_comm
    FROM EMPLOYEES
    WHERE EMPLOYEE_ID = v_empno;

    IF v_comm IS NULL THEN
        UPDATE EMPLOYEES
        SET SALARY = SALARY + 0
        WHERE EMPLOYEE_ID = v_empno;

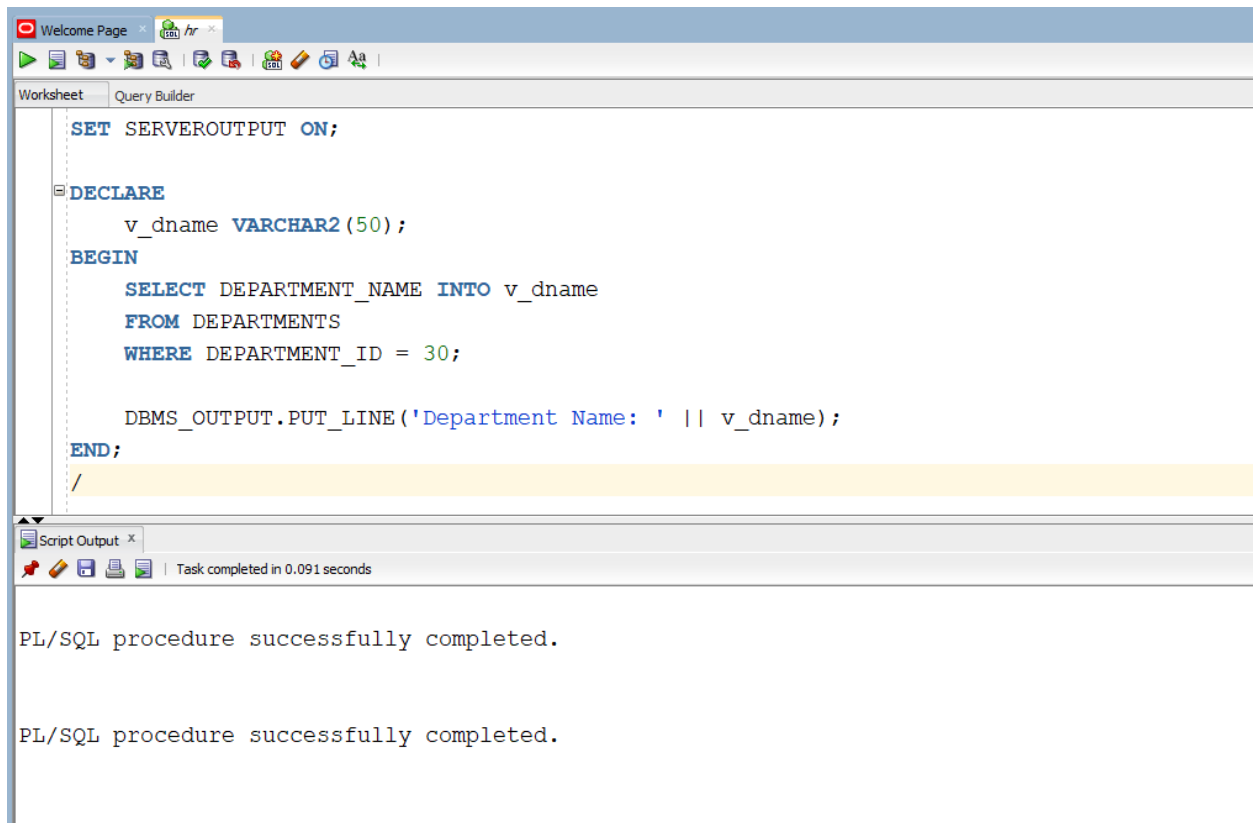
        DBMS_OUTPUT.PUT_LINE('Commission is NULL');
    ELSE
        UPDATE EMPLOYEES
        SET SALARY = SALARY + v_comm
        WHERE EMPLOYEE_ID = v_empno;

        DBMS_OUTPUT.PUT_LINE('Commission added');
    END IF;

    COMMIT;
```

Script Output x
Task completed in 0.115 seconds
PL/SQL procedure successfully completed.

Task 3:



The screenshot displays the Oracle SQL Developer environment. The main window is titled 'Worksheet' and contains a PL/SQL script. The script is as follows:

```
SET SERVEROUTPUT ON;

DECLARE
    v_dname VARCHAR2(50);
BEGIN
    SELECT DEPARTMENT_NAME INTO v_dname
    FROM DEPARTMENTS
    WHERE DEPARTMENT_ID = 30;

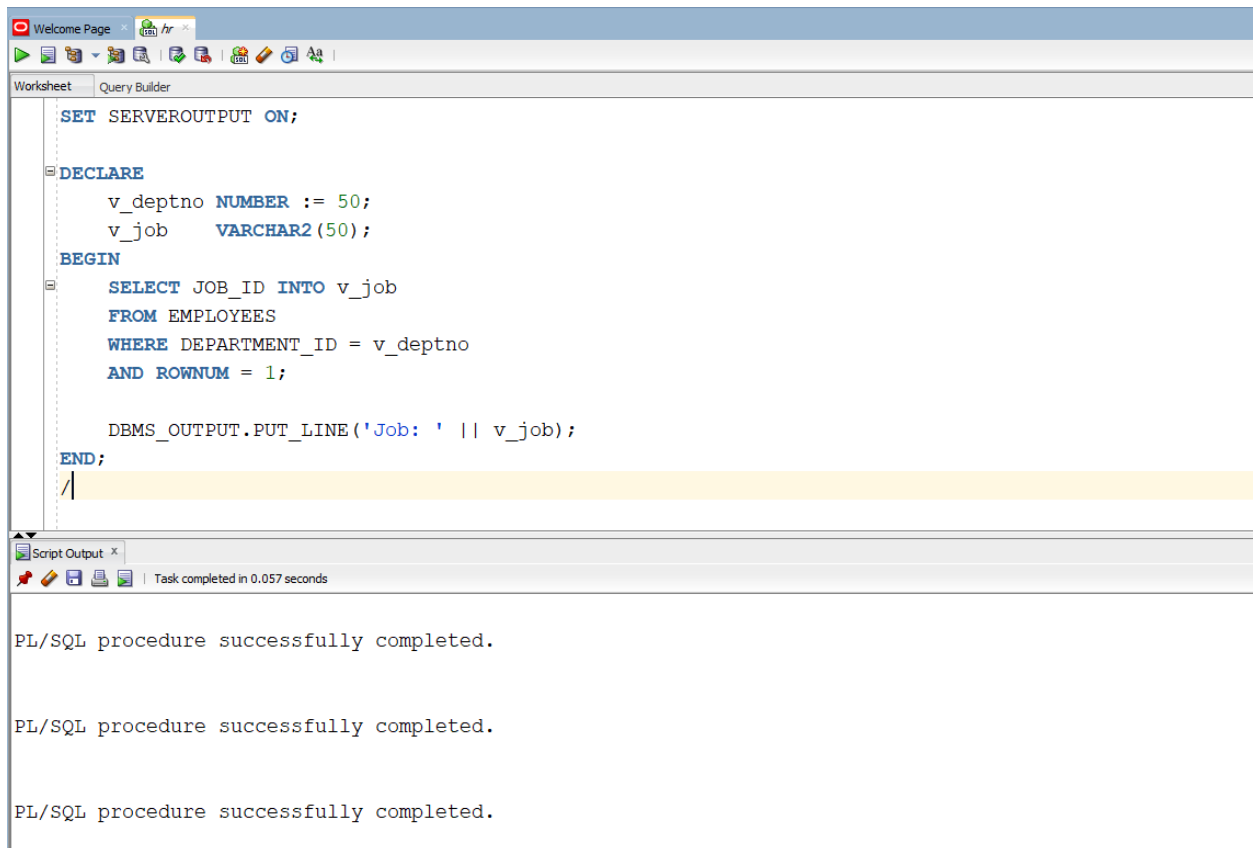
    DBMS_OUTPUT.PUT_LINE('Department Name: ' || v_dname);
END;
```

Below the script editor, the 'Script Output' window is visible, showing the message: 'Task completed in 0.091 seconds'. The output area displays the following text:

```
PL/SQL procedure successfully completed.
```

```
PL/SQL procedure successfully completed.
```

Task 4:



The screenshot displays the Oracle SQL Developer environment. The top window, titled 'Query Builder', contains the following PL/SQL script:

```
SET SERVEROUTPUT ON;

DECLARE
    v_deptno NUMBER := 50;
    v_job     VARCHAR2(50);
BEGIN
    SELECT JOB_ID INTO v_job
    FROM EMPLOYEES
    WHERE DEPARTMENT_ID = v_deptno
    AND ROWNUM = 1;

    DBMS_OUTPUT.PUT_LINE('Job: ' || v_job);
END;
/
```

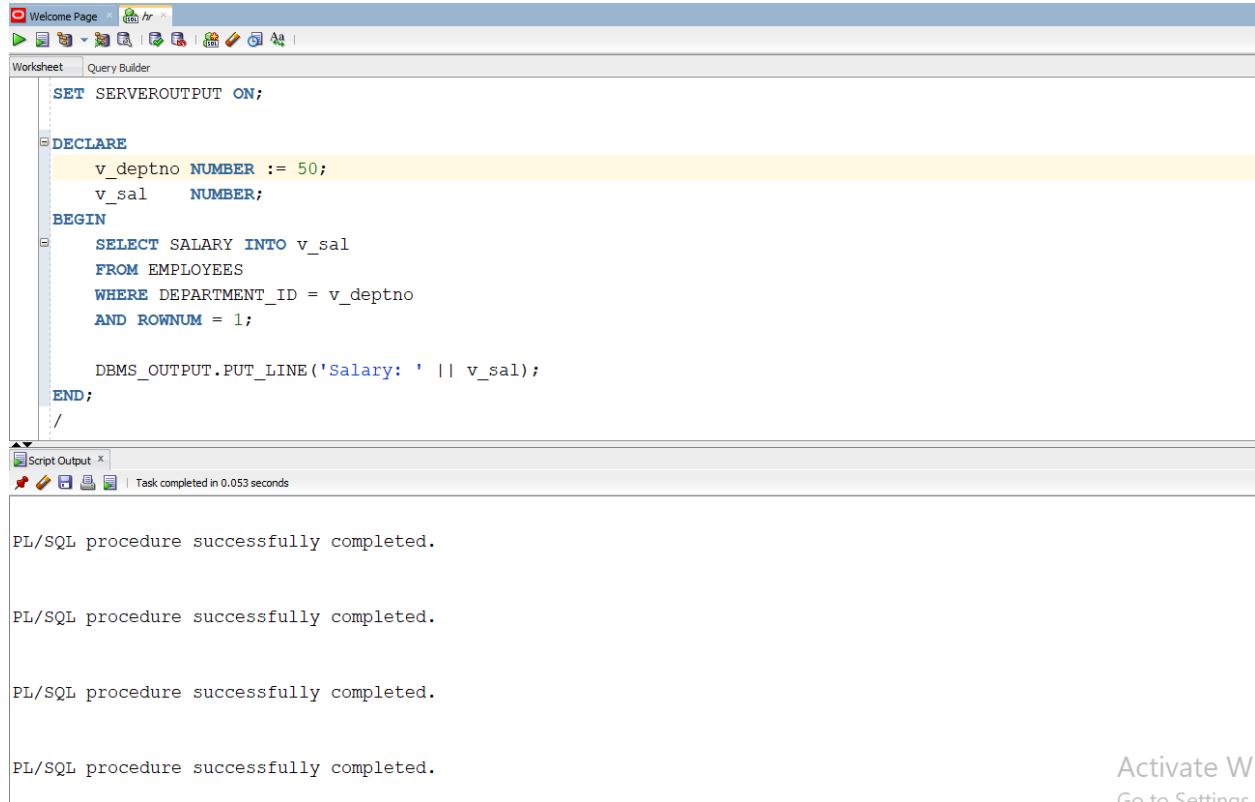
The bottom window, titled 'Script Output', shows the results of the script execution:

```
PL/SQL procedure successfully completed.

PL/SQL procedure successfully completed.

PL/SQL procedure successfully completed.
```

Task 5:



The screenshot displays the Oracle SQL Developer environment. The top window, titled 'Query Builder', contains a PL/SQL script. The script starts with 'SET SERVEROUTPUT ON;', followed by a 'DECLARE' section where a variable 'v_deptno' is set to 50 and 'v_sal' is declared as a NUMBER. The 'BEGIN' section contains a 'SELECT' statement that retrieves the salary for the department with ID 50, limited to one row. The result is then printed using 'DBMS_OUTPUT.PUT_LINE'. The script ends with 'END;' and a forward slash. Below the script editor, the 'Script Output' window shows the execution results, which consist of four identical messages: 'PL/SQL procedure successfully completed.'.

```
SET SERVEROUTPUT ON;

DECLARE
    v_deptno NUMBER := 50;
    v_sal     NUMBER;
BEGIN
    SELECT SALARY INTO v_sal
    FROM EMPLOYEES
    WHERE DEPARTMENT_ID = v_deptno
    AND ROWNUM = 1;

    DBMS_OUTPUT.PUT_LINE('Salary: ' || v_sal);
END;
/
```

Script Output x

Task completed in 0.053 seconds

PL/SQL procedure successfully completed.

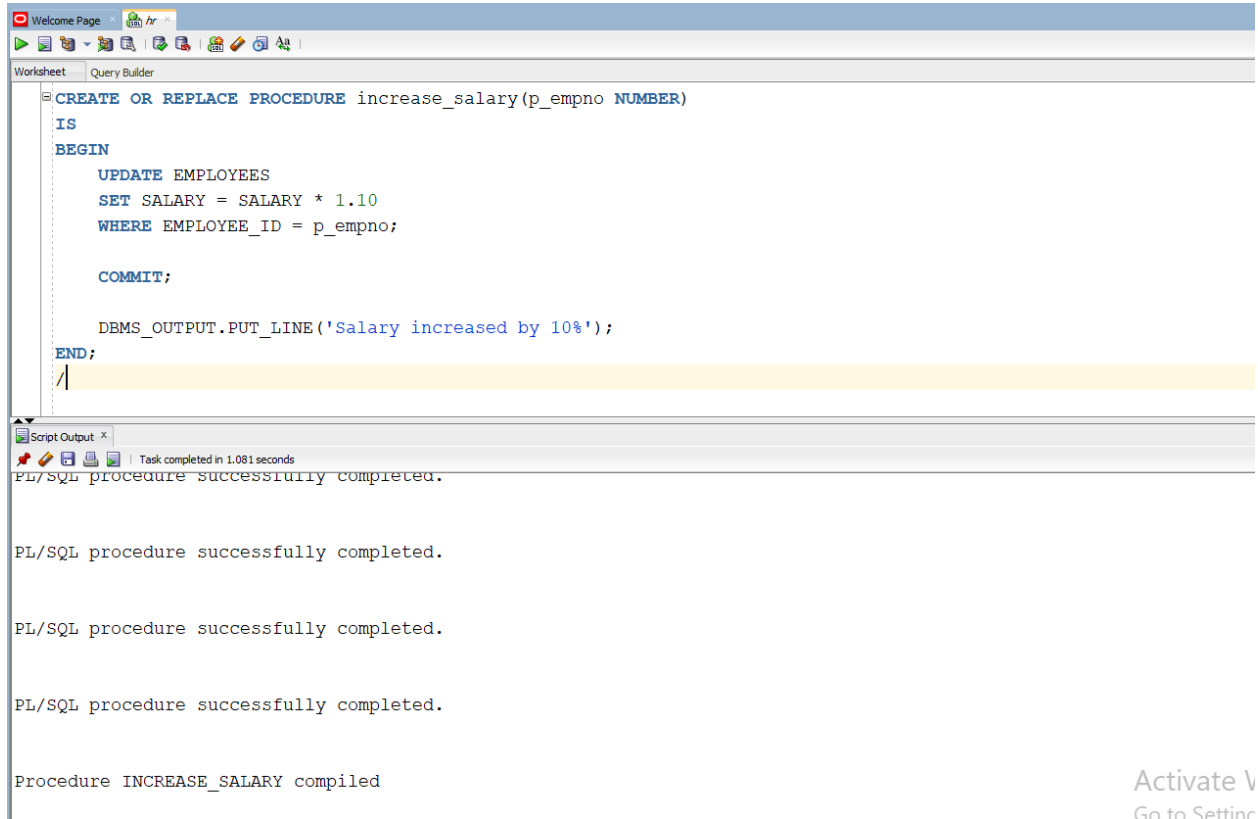
PL/SQL procedure successfully completed.

PL/SQL procedure successfully completed.

PL/SQL procedure successfully completed.

Activate Wi
Go to Settings

Task 6:



The screenshot shows the SQL Developer interface with the 'Query Builder' tab active. The SQL editor contains the following code:

```
CREATE OR REPLACE PROCEDURE increase_salary(p_empno NUMBER)
IS
BEGIN
    UPDATE EMPLOYEES
    SET SALARY = SALARY * 1.10
    WHERE EMPLOYEE_ID = p_empno;

    COMMIT;

    DBMS_OUTPUT.PUT_LINE('Salary increased by 10%');
END;
/
```

The 'Script Output' window at the bottom shows the execution results:

```
Task completed in 1.081 seconds
PL/SQL procedure successfully completed.

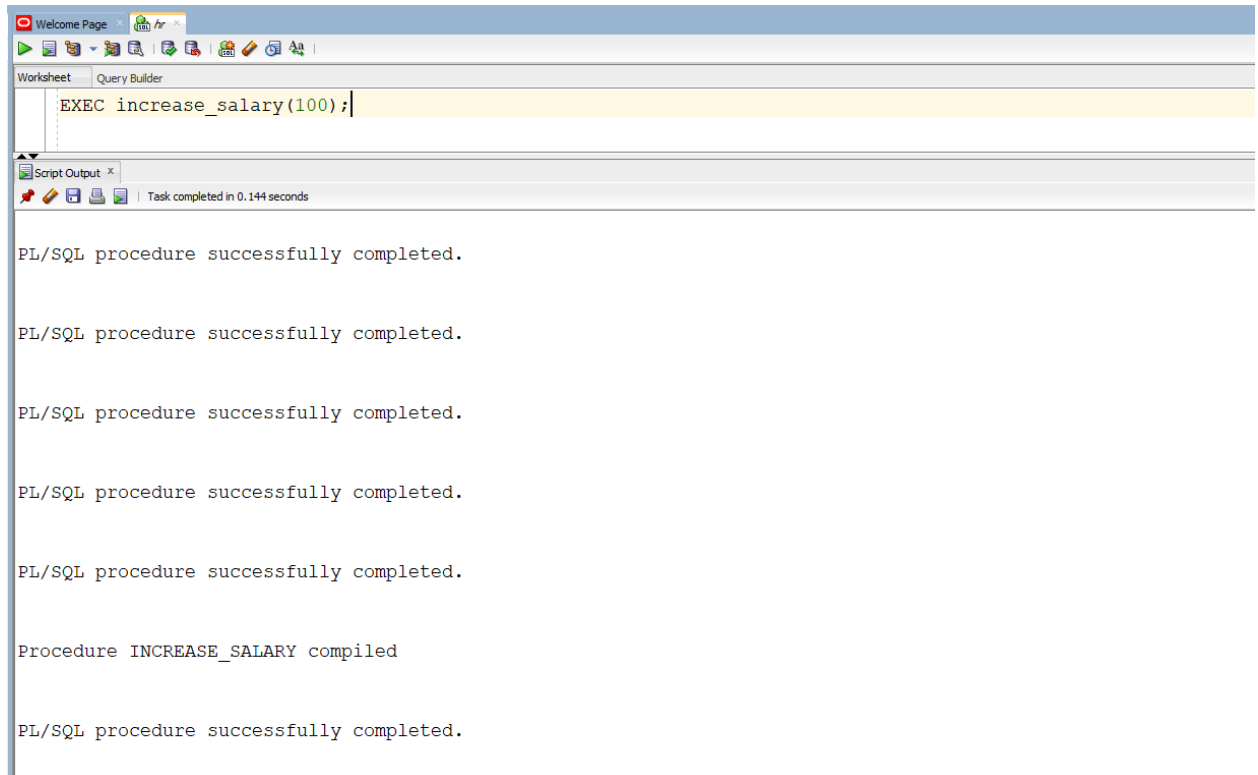
PL/SQL procedure successfully completed.

PL/SQL procedure successfully completed.

PL/SQL procedure successfully completed.

Procedure INCREASE_SALARY compiled
```

On the right side of the interface, there is a watermark that reads: 'Activate V Go to Setting'.



The screenshot shows the SQL Developer interface with the 'Query Builder' tab active. The SQL editor contains the following code:

```
EXEC increase_salary(100);
```

The 'Script Output' window at the bottom shows the execution results:

```
Task completed in 0.144 seconds

PL/SQL procedure successfully completed.

PL/SQL procedure successfully completed.

PL/SQL procedure successfully completed.

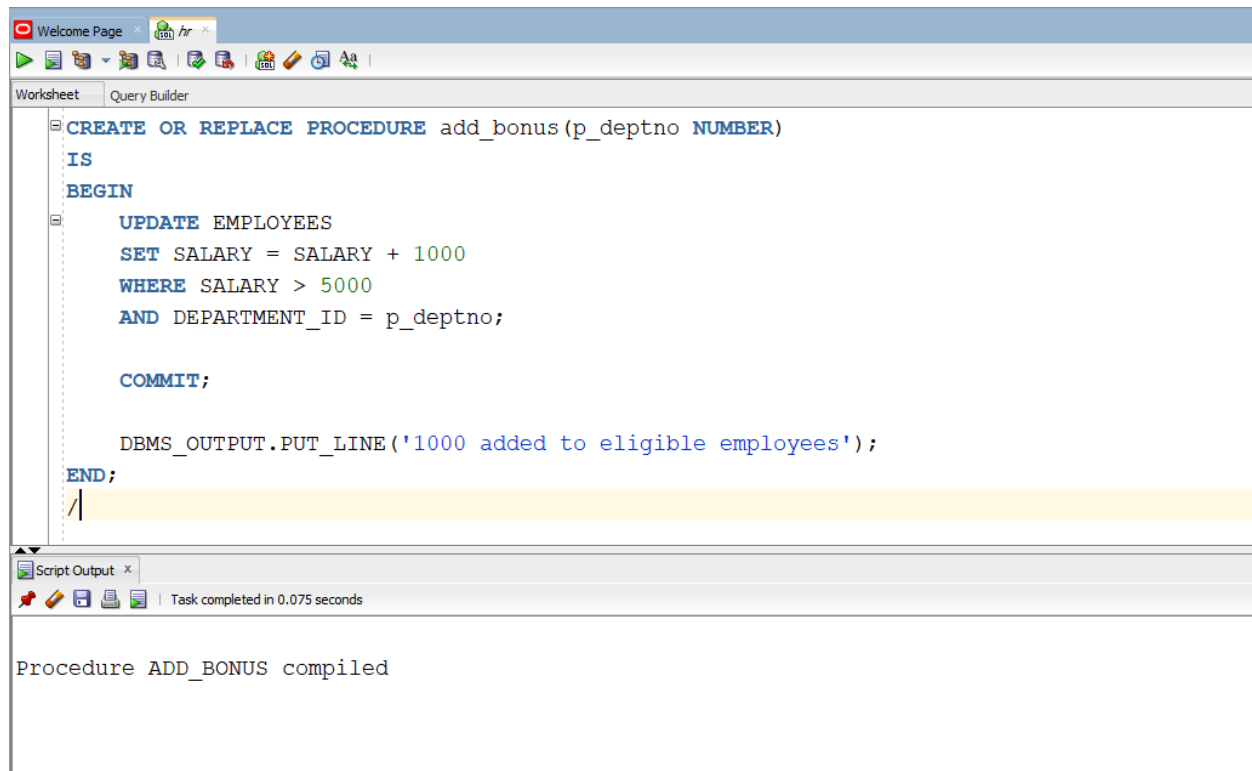
PL/SQL procedure successfully completed.

PL/SQL procedure successfully completed.

Procedure INCREASE_SALARY compiled

PL/SQL procedure successfully completed.
```

Task 7:



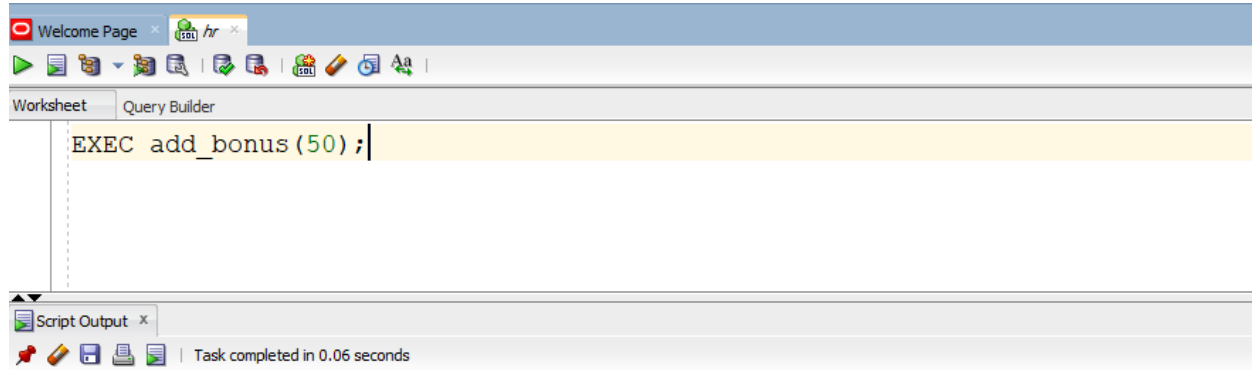
The screenshot shows the SQL Developer interface with the Query Builder tab active. The SQL editor contains the following code:

```
CREATE OR REPLACE PROCEDURE add_bonus(p_deptno NUMBER)
IS
BEGIN
    UPDATE EMPLOYEES
    SET SALARY = SALARY + 1000
    WHERE SALARY > 5000
    AND DEPARTMENT_ID = p_deptno;

    COMMIT;

    DBMS_OUTPUT.PUT_LINE('1000 added to eligible employees');
END;
```

The Script Output window at the bottom shows the message: "Task completed in 0.075 seconds". Below the output window, the text "Procedure ADD_BONUS compiled" is displayed.



The screenshot shows the SQL Developer interface with the Query Builder tab active. The SQL editor contains the following code:

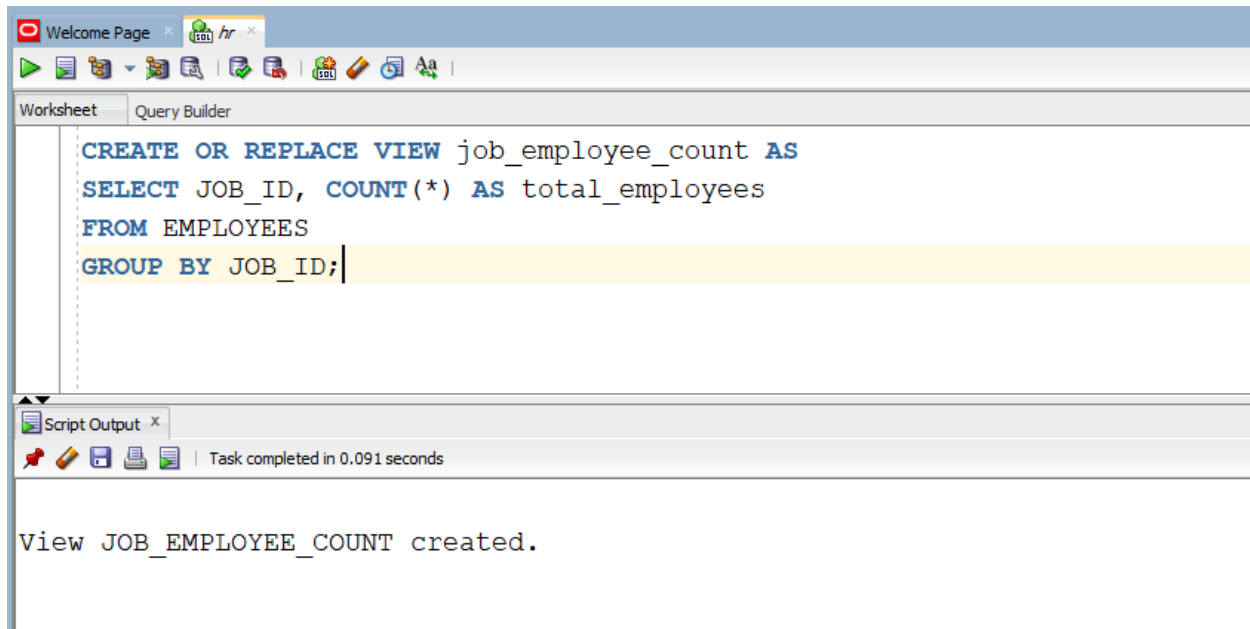
```
EXEC add_bonus(50);
```

The Script Output window at the bottom shows the message: "Task completed in 0.06 seconds". Below the output window, the text "Procedure ADD_BONUS compiled" is displayed.

PL/SQL procedure successfully completed.

Post Lab Tasks

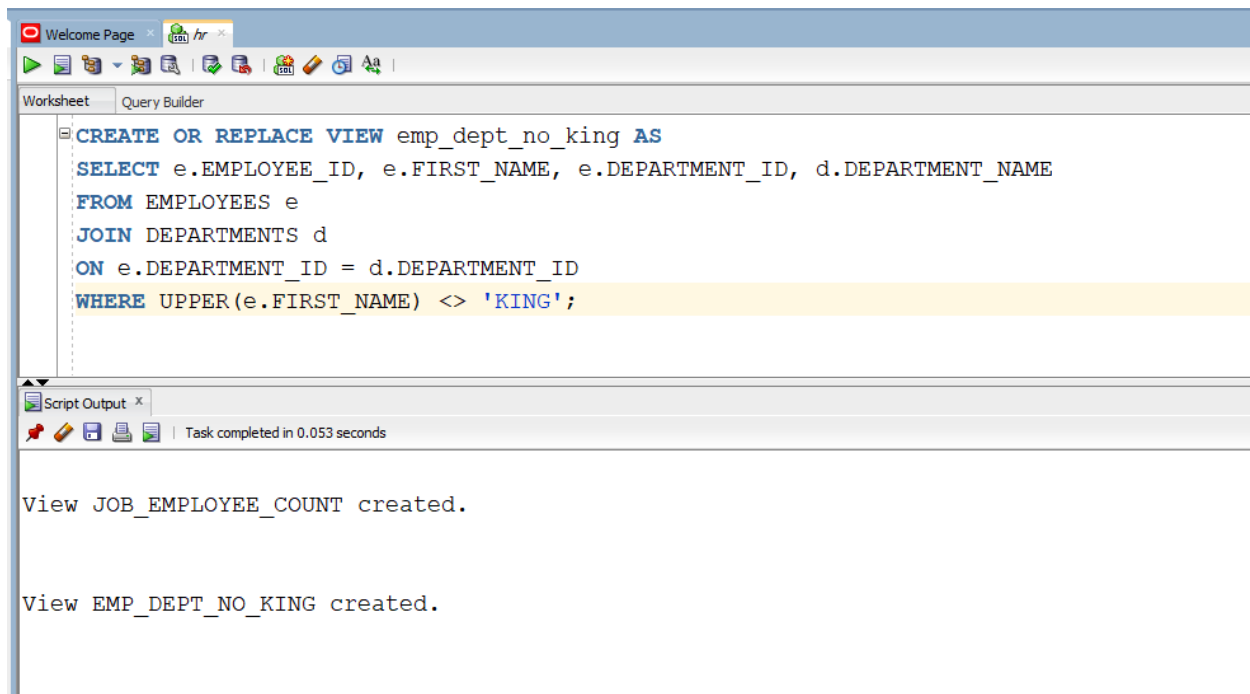
Task 1:



The screenshot shows the SQL Developer interface with a 'Query Builder' tab. The SQL editor contains the following code:

```
CREATE OR REPLACE VIEW job_employee_count AS
SELECT JOB_ID, COUNT(*) AS total_employees
FROM EMPLOYEES
GROUP BY JOB_ID;
```

Below the editor, the 'Script Output' tab shows the message: 'Task completed in 0.091 seconds'. Below the output, the text 'View JOB_EMPLOYEE_COUNT created.' is displayed.



The screenshot shows the SQL Developer interface with a 'Query Builder' tab. The SQL editor contains the following code:

```
CREATE OR REPLACE VIEW emp_dept_no_king AS
SELECT e.EMPLOYEE_ID, e.FIRST_NAME, e.DEPARTMENT_ID, d.DEPARTMENT_NAME
FROM EMPLOYEES e
JOIN DEPARTMENTS d
ON e.DEPARTMENT_ID = d.DEPARTMENT_ID
WHERE UPPER(e.FIRST_NAME) <> 'KING';
```

Below the editor, the 'Script Output' tab shows the message: 'Task completed in 0.053 seconds'. Below the output, the text 'View JOB_EMPLOYEE_COUNT created.' is displayed, followed by 'View EMP_DEPT_NO_KING created.'

The screenshot shows a database query tool interface. At the top, there are tabs for 'Welcome Page' and 'hr'. Below the tabs is a toolbar with various icons. The main area is divided into two panes: 'Worksheet' and 'Query Builder'. The 'Query Builder' pane contains the following SQL query:

```
CREATE OR REPLACE VIEW emp_dept_all AS
SELECT e.EMPLOYEE_ID, e.FIRST_NAME, e.DEPARTMENT_ID, d.DEPARTMENT_NAME
FROM EMPLOYEES e
JOIN DEPARTMENTS d
ON e.DEPARTMENT_ID = d.DEPARTMENT_ID;
```

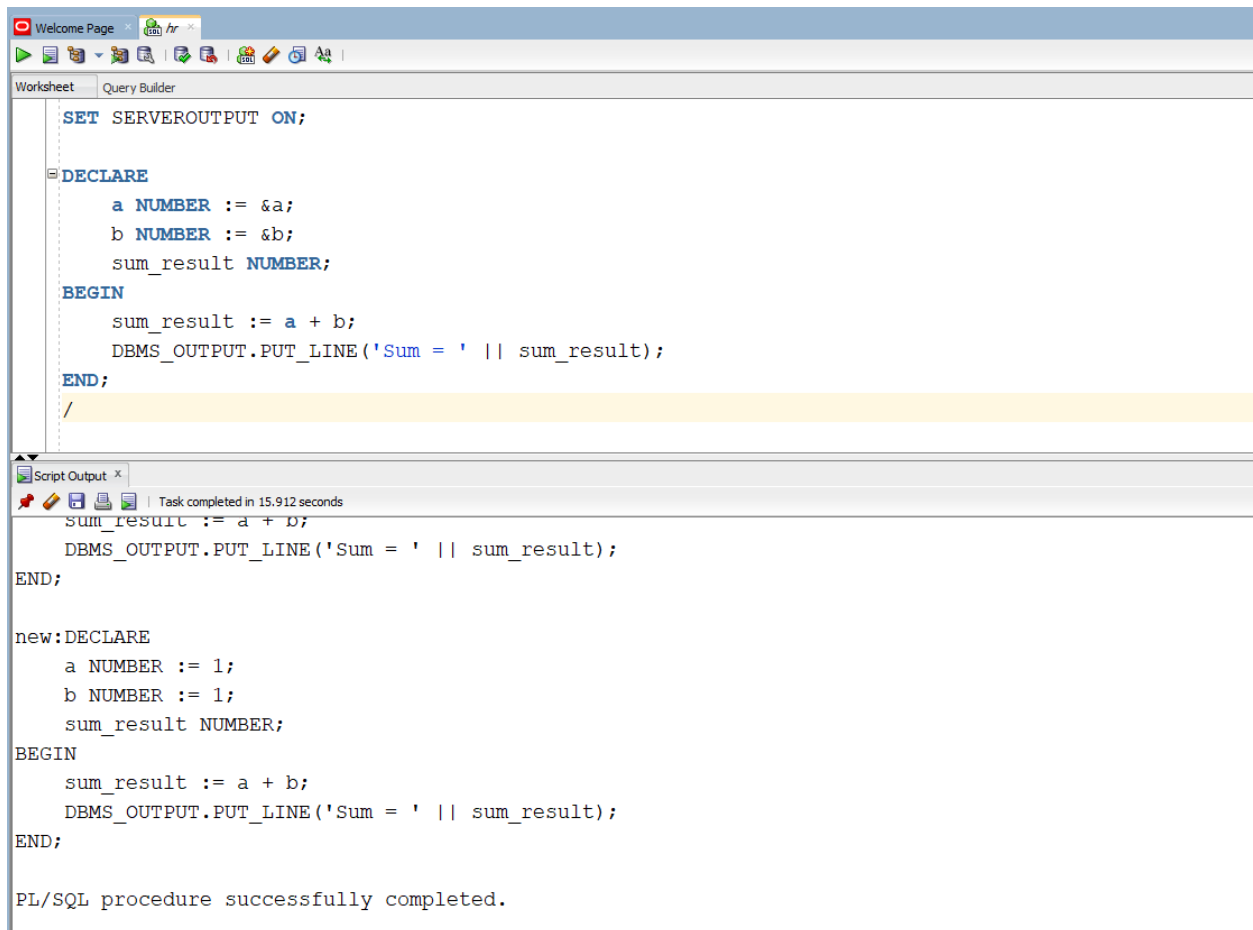
Below the query pane is a 'Script Output' pane. It shows the results of the query execution, which are three messages indicating the creation of views:

```
View JOB_EMPLOYEE_COUNT created.

View EMP_DEPT_NO_KING created.

View EMP_DEPT_ALL created.
```

Task 2:



The screenshot displays the Oracle SQL Developer environment. The top pane, titled 'Query Builder', contains a PL/SQL script. The script starts with 'SET SERVEROUTPUT ON;', followed by a 'DECLARE' block where variables 'a' and 'b' are declared as 'NUMBER' and assigned values '&a;' and '&b;' respectively. A 'sum_result' variable is also declared as 'NUMBER'. The 'BEGIN' block contains the calculation 'sum_result := a + b;' and a call to 'DBMS_OUTPUT.PUT_LINE('Sum = ' || sum_result);'. The script ends with 'END;' and a forward slash '/' on a highlighted yellow line. The bottom pane, titled 'Script Output', shows the execution results. It indicates 'Task completed in 15.912 seconds' and displays the output of the script, which is identical to the code in the top pane. The output concludes with the message 'PL/SQL procedure successfully completed.'

```
SET SERVEROUTPUT ON;

DECLARE
    a NUMBER := &a;
    b NUMBER := &b;
    sum_result NUMBER;
BEGIN
    sum_result := a + b;
    DBMS_OUTPUT.PUT_LINE('Sum = ' || sum_result);
END;
/
```

Script Output x

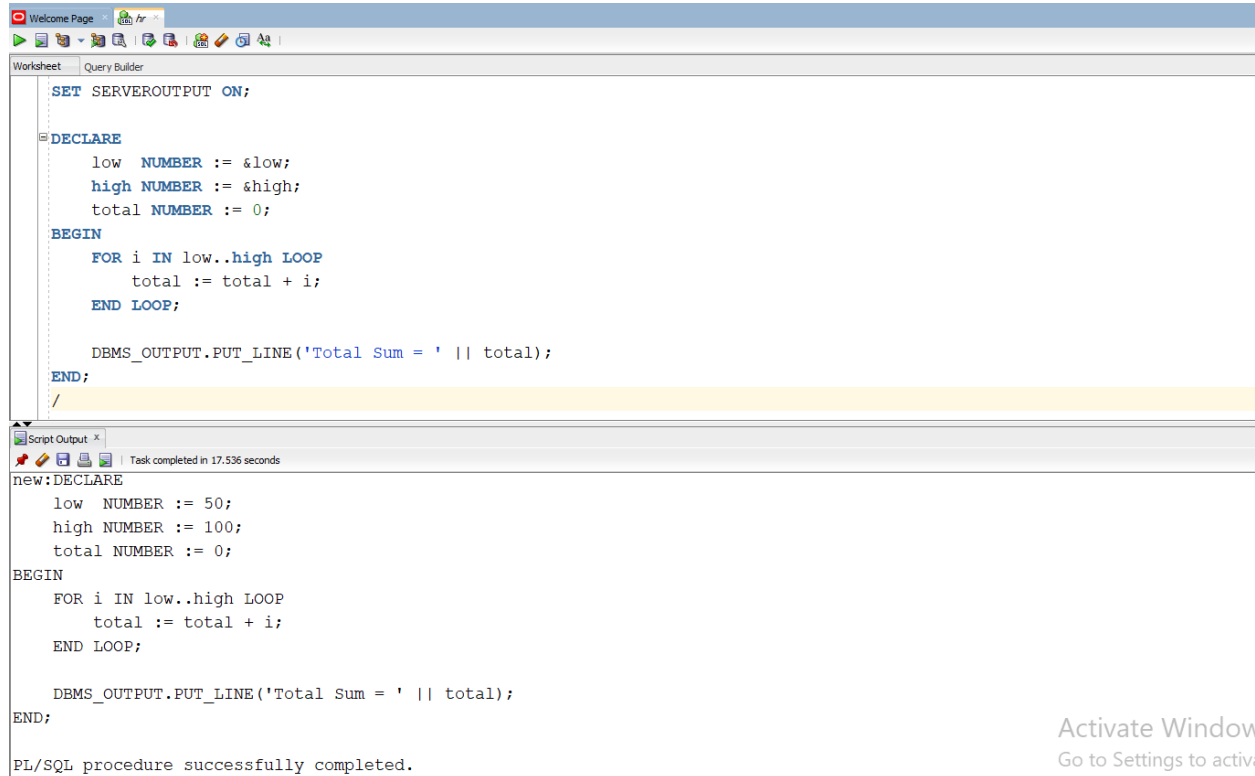
Task completed in 15.912 seconds

```
sum_result := a + b;
DBMS_OUTPUT.PUT_LINE('Sum = ' || sum_result);
END;

new:DECLARE
    a NUMBER := 1;
    b NUMBER := 1;
    sum_result NUMBER;
BEGIN
    sum_result := a + b;
    DBMS_OUTPUT.PUT_LINE('Sum = ' || sum_result);
END;

PL/SQL procedure successfully completed.
```

Task 3:



The screenshot displays the Oracle SQL Developer environment. The top toolbar includes icons for running, saving, and other database operations. The 'Worksheet' tab is active, showing a PL/SQL script. The script starts with `SET SERVEROUTPUT ON;`, followed by a `DECLARE` block where variables `low`, `high`, and `total` are initialized. A `BEGIN` block contains a `FOR` loop that iterates from `low` to `high`, incrementing `total` by `i` in each iteration. After the loop, the script uses `DBMS_OUTPUT.PUT_LINE` to display the total sum, followed by `END;` and a forward slash to execute the block.

The 'Script Output' window at the bottom shows the execution results. It indicates that the task was completed in 17.536 seconds. The output text is identical to the script code, showing the variable declarations, the loop execution, and the final output line: `DBMS_OUTPUT.PUT_LINE('Total Sum = ' || total);`. Below the output, a message states: 'PL/SQL procedure successfully completed.'

```
SET SERVEROUTPUT ON;

DECLARE
    low  NUMBER := &low;
    high NUMBER := &high;
    total NUMBER := 0;
BEGIN
    FOR i IN low..high LOOP
        total := total + i;
    END LOOP;

    DBMS_OUTPUT.PUT_LINE('Total Sum = ' || total);
END;
/
```

Script Output x
Task completed in 17.536 seconds

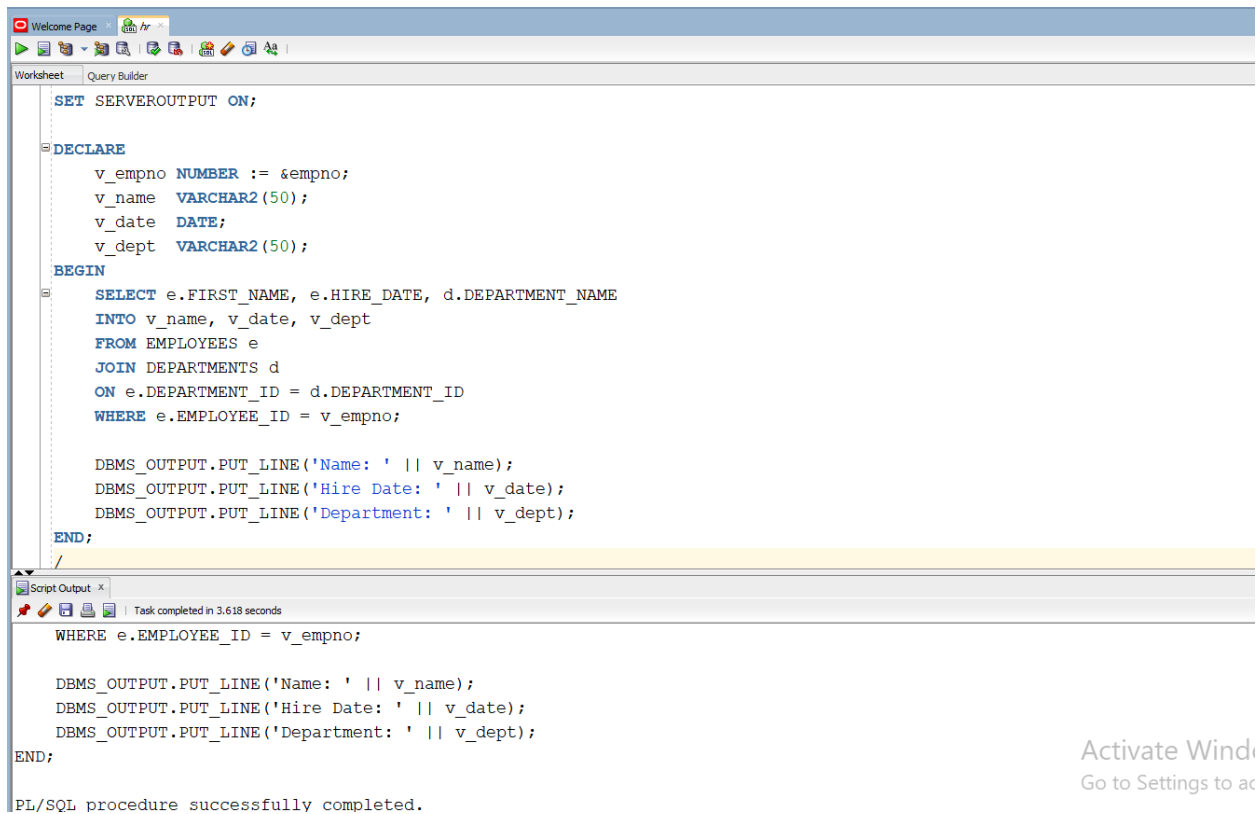
```
new:DECLARE
    low  NUMBER := 50;
    high NUMBER := 100;
    total NUMBER := 0;
BEGIN
    FOR i IN low..high LOOP
        total := total + i;
    END LOOP;

    DBMS_OUTPUT.PUT_LINE('Total Sum = ' || total);
END;
```

PL/SQL procedure successfully completed.

Activate Window
Go to Settings to activate

Task 4:



The screenshot displays the Oracle SQL Developer environment. The main window shows a PL/SQL procedure script in the 'Query Builder' tab. The script is as follows:

```
SET SERVEROUTPUT ON;

DECLARE
    v_empno NUMBER := &empno;
    v_name VARCHAR2(50);
    v_date DATE;
    v_dept VARCHAR2(50);
BEGIN
    SELECT e.FIRST_NAME, e.HIRE_DATE, d.DEPARTMENT_NAME
    INTO v_name, v_date, v_dept
    FROM EMPLOYEES e
    JOIN DEPARTMENTS d
    ON e.DEPARTMENT_ID = d.DEPARTMENT_ID
    WHERE e.EMPLOYEE_ID = v_empno;

    DBMS_OUTPUT.PUT_LINE('Name: ' || v_name);
    DBMS_OUTPUT.PUT_LINE('Hire Date: ' || v_date);
    DBMS_OUTPUT.PUT_LINE('Department: ' || v_dept);
END;
```

Below the script, the 'Script Output' window shows the execution results:

```
Task completed in 3.618 seconds

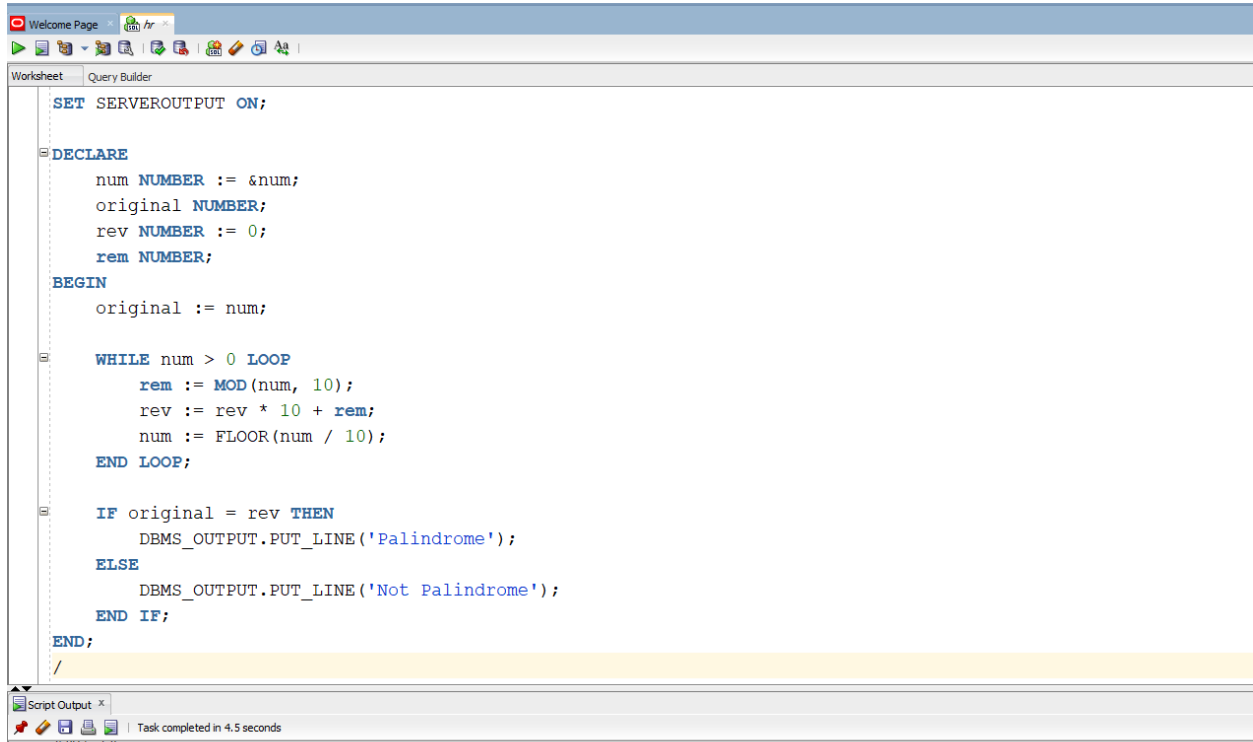
WHERE e.EMPLOYEE_ID = v_empno;

DBMS_OUTPUT.PUT_LINE('Name: ' || v_name);
DBMS_OUTPUT.PUT_LINE('Hire Date: ' || v_date);
DBMS_OUTPUT.PUT_LINE('Department: ' || v_dept);
END;
```

At the bottom of the interface, a status message reads: "PL/SQL procedure successfully completed."

On the right side of the image, there is a watermark text: "Activate Windows Go to Settings to activate Windows."

Task 5:



```
SET SERVEROUTPUT ON;

DECLARE
    num NUMBER := &num;
    original NUMBER;
    rev NUMBER := 0;
    rem NUMBER;
BEGIN
    original := num;

    WHILE num > 0 LOOP
        rem := MOD(num, 10);
        rev := rev * 10 + rem;
        num := FLOOR(num / 10);
    END LOOP;

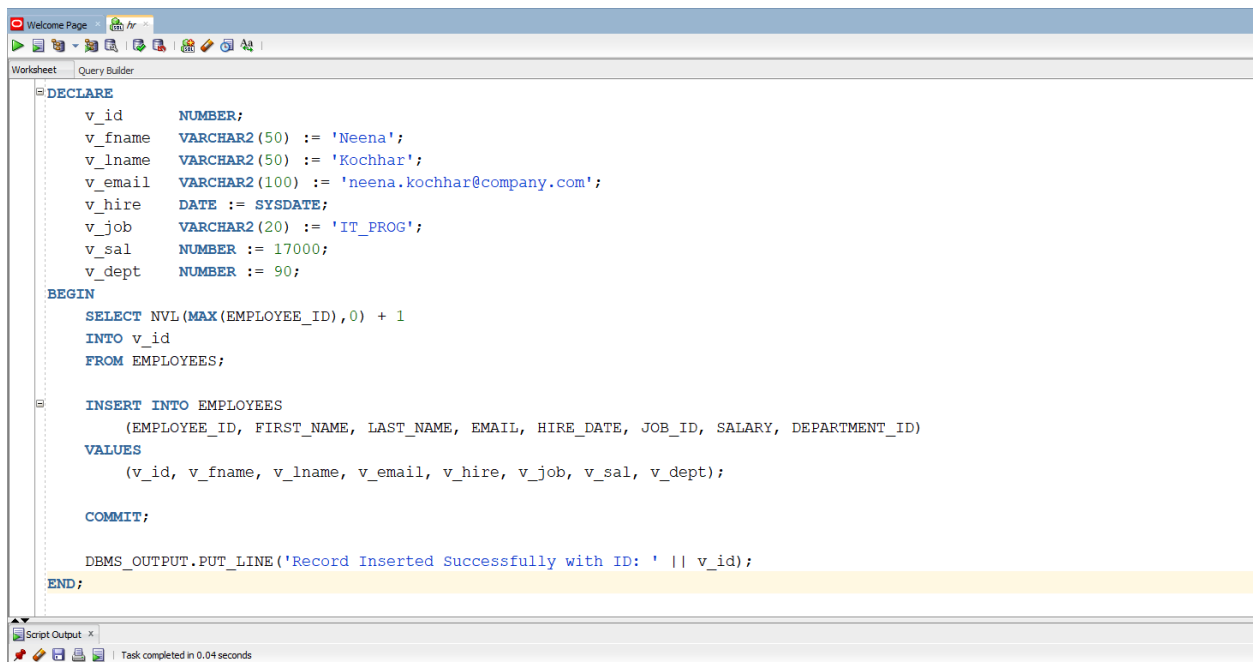
    IF original = rev THEN
        DBMS_OUTPUT.PUT_LINE('Palindrome');
    ELSE
        DBMS_OUTPUT.PUT_LINE('Not Palindrome');
    END IF;
END;
/
```

END;

PL/SQL procedure successfully completed.

Activate Windows
Go to Settings to activate Windows

Task 6:



```
DECLARE
    v_id      NUMBER;
    v_fname   VARCHAR2(50) := 'Neena';
    v_lname   VARCHAR2(50) := 'Kochhar';
    v_email   VARCHAR2(100) := 'neena.kochhar@company.com';
    v_hire     DATE := SYSDATE;
    v_job      VARCHAR2(20) := 'IT_PROG';
    v_sal      NUMBER := 17000;
    v_dept     NUMBER := 90;
BEGIN
    SELECT NVL(MAX(EMPLOYEE_ID), 0) + 1
    INTO v_id
    FROM EMPLOYEES;

    INSERT INTO EMPLOYEES
        (EMPLOYEE_ID, FIRST_NAME, LAST_NAME, EMAIL, HIRE_DATE, JOB_ID, SALARY, DEPARTMENT_ID)
    VALUES
        (v_id, v_fname, v_lname, v_email, v_hire, v_job, v_sal, v_dept);

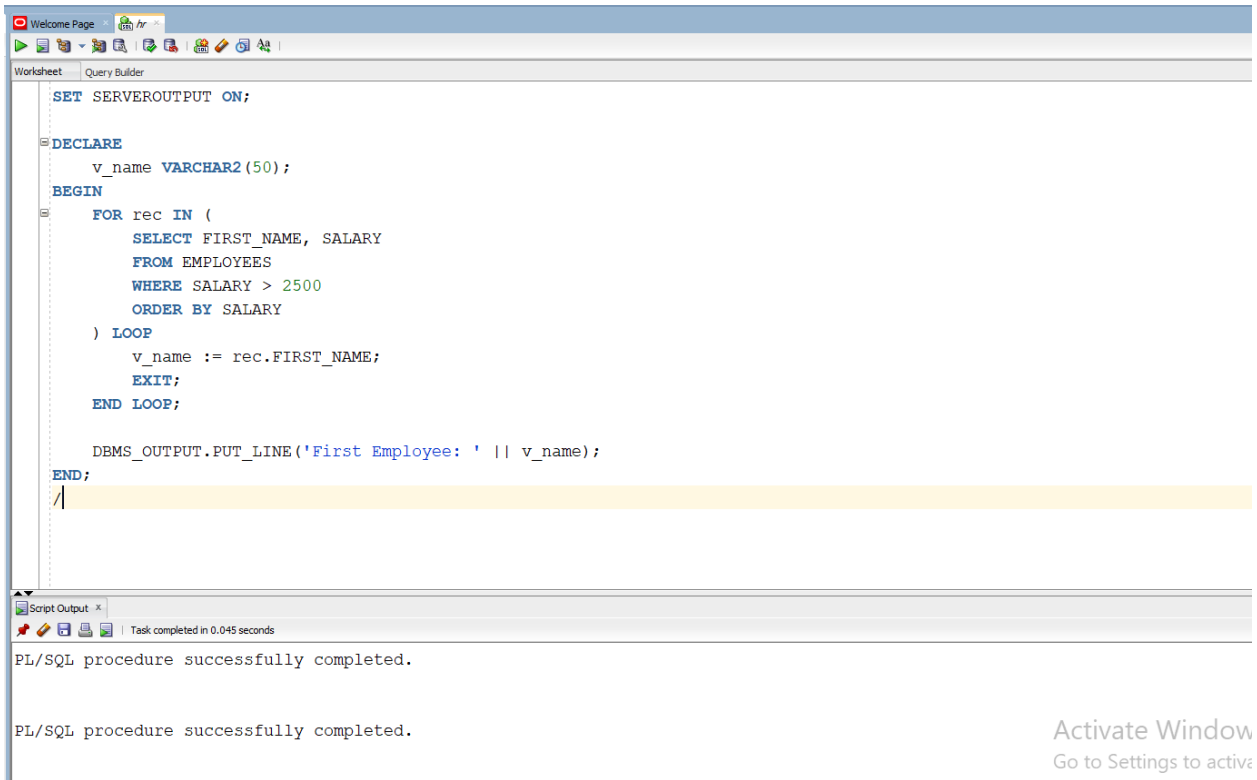
    COMMIT;

    DBMS_OUTPUT.PUT_LINE('Record Inserted Successfully with ID: ' || v_id);
END;
```

PL/SQL procedure successfully completed.

Activate Windows
Go to Settings to activate Windows

Task 7:



The screenshot shows the SQL Developer interface with a script editor containing the following PL/SQL code:

```
SET SERVEROUTPUT ON;

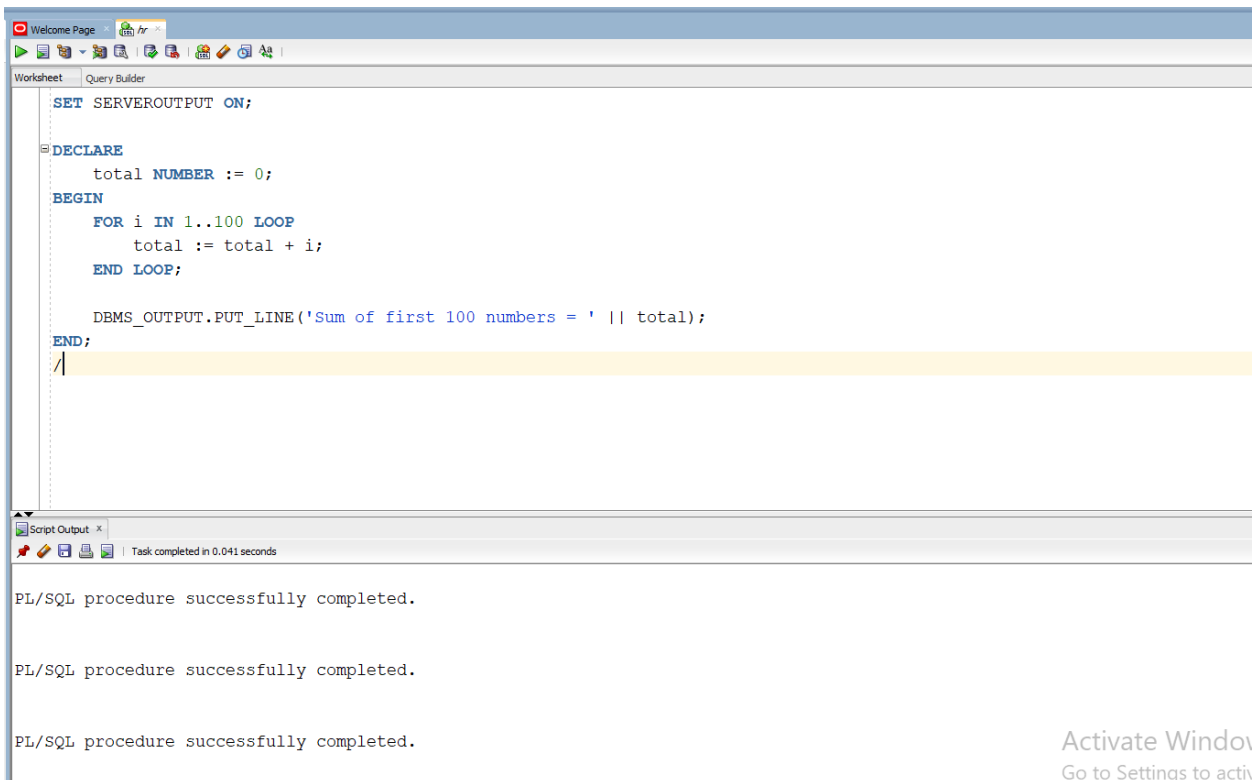
DECLARE
    v_name VARCHAR2(50);
BEGIN
    FOR rec IN (
        SELECT FIRST_NAME, SALARY
        FROM EMPLOYEES
        WHERE SALARY > 2500
        ORDER BY SALARY
    ) LOOP
        v_name := rec.FIRST_NAME;
        EXIT;
    END LOOP;

    DBMS_OUTPUT.PUT_LINE('First Employee: ' || v_name);
END;
/
```

Below the script editor, the 'Script Output' window shows the message: 'PL/SQL procedure successfully completed.'.

Activate Window
Go to Settings to activate

Task 8:



The screenshot shows the SQL Developer interface with a script editor containing the following PL/SQL code:

```
SET SERVEROUTPUT ON;

DECLARE
    total NUMBER := 0;
BEGIN
    FOR i IN 1..100 LOOP
        total := total + i;
    END LOOP;

    DBMS_OUTPUT.PUT_LINE('Sum of first 100 numbers = ' || total);
END;
/
```

Below the script editor, the 'Script Output' window shows the message: 'PL/SQL procedure successfully completed.'.

Activate Window
Go to Settings to activate